

Setting Up Visual Studio Code (VSCode)

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What is an IDE?

- IDE == Integrated development environment

```
"""Provides an implementation of the classic Hello World function."""

a_new_variable = ""

def hello_world() -> str:
    """A simple function that returns 'Hello World!'

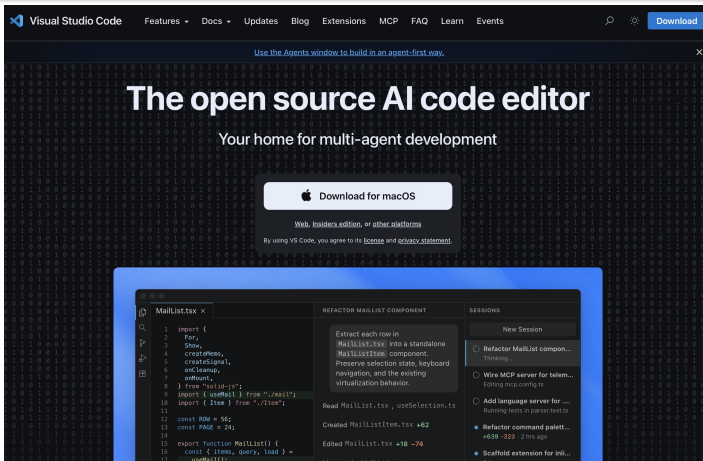
    :returns: String containing 'Hello World!'
    :rtype: str
    """

    return Hello World!"
```

```
1  """Provides an implementation of the classic Hello World function."""
2
3
4  a_new_variable = ""
5
6  def hello_world() -> str:
7      """A simple function that returns 'Hello World!'
8
9      :returns: String containing 'Hello World!'
10     :rtype: str
11     """
12
13     return Hello World!"  SyntaxError: unterminated string literal (detected at line 13)
14
```

Installing VSCode

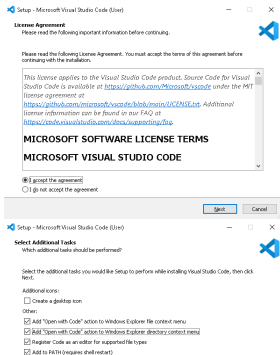
- Go to <https://code.visualstudio.com>
- Click “Download for Your_OS”
- Open the installer



Installing VSCode

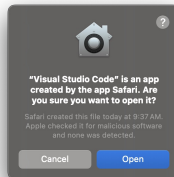
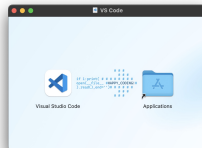
Windows

- Make sure all “Open with Code” options are checked
- Finish and launch VSCode



MacOS

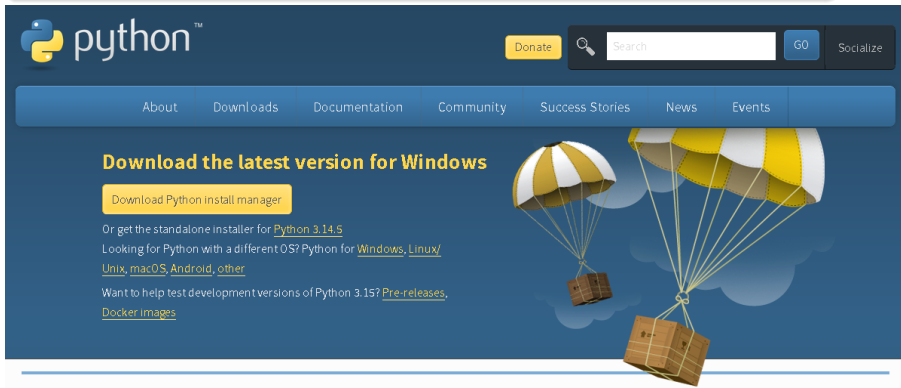
- Drag icon to “Applications”
- Open “Visual Studio Code” from Launchpad



- VSCode may prompt you to sign in to gain access to AI models/agents.
- Do so at your own risk.
- You are responsible for every line of code you commit to a repository.
- If you let AI develop for you, you must validate the results.
- Validation is nearly impossible unless you are experienced already.

Installing Python (Windows)

- Go to <https://python.org/downloads/>
- Click “Download Python install manager”
- Open the installer



The screenshot shows the Python.org website. At the top left is the Python logo. To its right is a yellow 'Donate' button, a search bar with a magnifying glass icon, and a 'GO' button. Further right is a 'Socialize' button. Below these is a navigation bar with links: 'About', 'Downloads', 'Documentation', 'Community', 'Success Stories', 'News', and 'Events'. The main content area has a heading 'Download the latest version for Windows' in yellow. Below it is a yellow button that says 'Download Python install manager'. To the right of this button is an illustration of two parachutes with yellow and white stripes, each carrying a brown cardboard box. Below the button, there is text: 'Or get the standalone installer for [Python 3.14.5](#)'. Then, 'Looking for Python with a different OS? Python for [Windows](#), [Linux/Unix](#), [macOS](#), [Android](#), [other](#)'. Finally, 'Want to help test development versions of Python 3.15? [Pre-releases](#), [Docker Images](#)'.

python™

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Search

GO

Socialize

About Downloads Documentation Community Success Stories News Events

Download the latest version for Windows

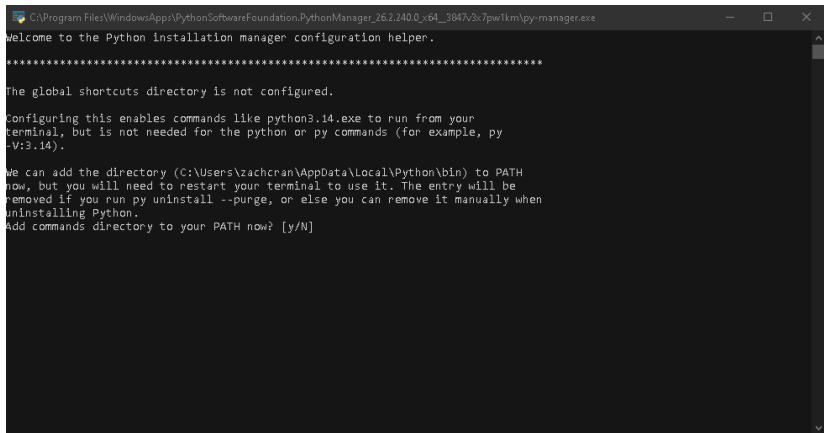
Download Python install manager

Or get the standalone installer for [Python 3.14.5](#)

Looking for Python with a different OS? Python for [Windows](#), [Linux/Unix](#), [macOS](#), [Android](#), [other](#)

Want to help test development versions of Python 3.15? [Pre-releases](#), [Docker Images](#)

Installing Python (Windows)

A screenshot of a Windows command prompt window titled "C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_26.2.240.0_x64_3847v3x7pw1km\py-manager.exe". The window displays the "Welcome to the Python installation manager configuration helper." message, followed by a separator line of asterisks. It then informs the user that the global shortcuts directory is not configured. The text explains that configuring this allows commands like python3.14.exe to run from the terminal, but it's not needed for python or py commands. It then offers to add the directory C:\Users\zachcran\AppData\Local\Python\bin to the PATH, noting that the entry will be removed if the user runs py uninstall --purge or manually when uninstalling Python. Finally, it asks "Add commands directory to your PATH now? [y/N]".

```
C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_26.2.240.0_x64_3847v3x7pw1km\py-manager.exe
Welcome to the Python installation manager configuration helper.

*****

The global shortcuts directory is not configured.

Configuring this enables commands like python3.14.exe to run from your
terminal, but is not needed for the python or py commands (for example, py
-V:3.14).

We can add the directory (C:\Users\zachcran\AppData\Local\Python\bin) to PATH
now, but you will need to restart your terminal to use it. The entry will be
removed if you run py uninstall --purge, or else you can remove it manually when
uninstalling Python.
Add commands directory to your PATH now? [y/N]
```

Installing Python (Windows)

```
*****  
You do not have the latest Python runtime.  
  
Install the current latest version of CPython? If not, you can use 'py install  
default' later to install.  
  
Install CPython now? [Y/n]
```


Testing your Python Install

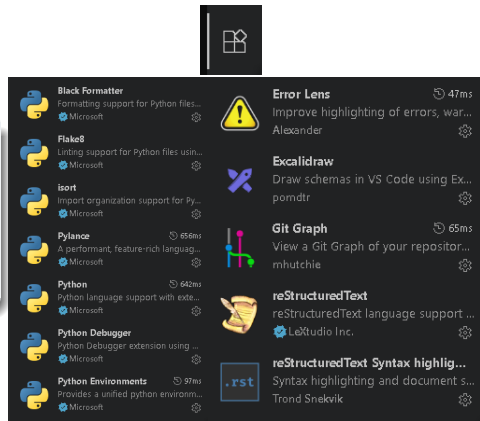
- Open VSCode
- At the top, select “Terminal”, “New Terminal”
- Test the **interpreter**: Type “python” and press [Enter]

```
C:\Users\zachcran>python
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello World!")
Hello World!
>>>
```

- Note: The interpreter is a place to quickly test individual functions and small scripts, but you should not develop code here!!!

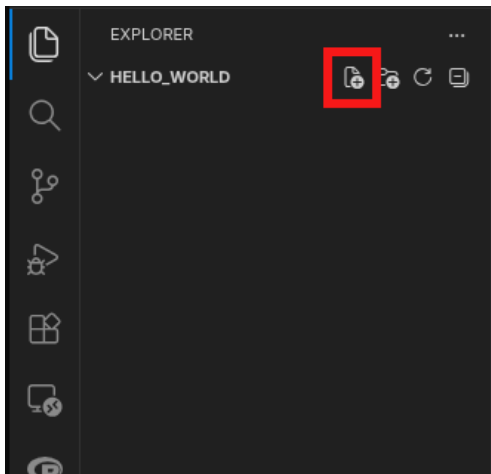
Installing Extensions

- Search for “Python” in the extensions tab
- Install Python extensions (left)
- Install recommended extensions (right)



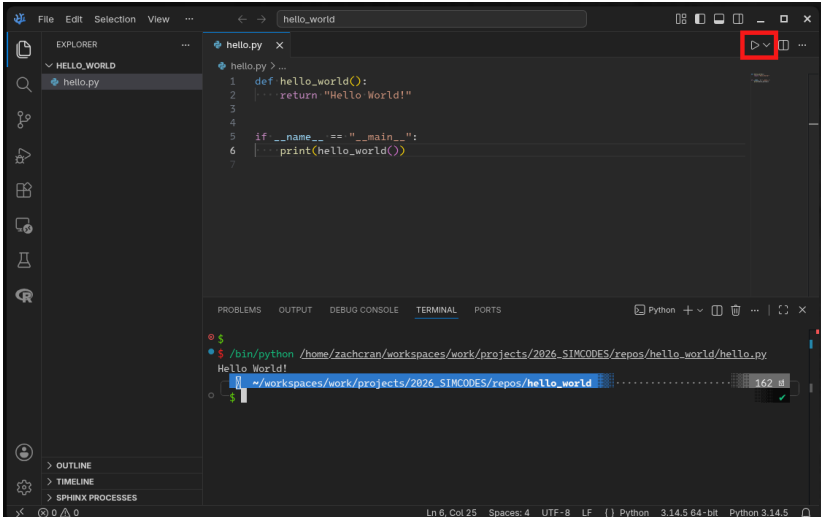
Testing your Development Environment

- Make a new folder named “hello_world”
- Open VSCode in that folder
- In VSCode, make a new file called “hello.py”



Testing your Development Environment

- Create and run the following code for a “Hello World” script



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named 'HELLO_WORLD' with a file 'hello.py'. The main editor window displays the contents of 'hello.py', which contains the following Python code:

```
1 def hello_world():  
2     return "Hello World!"  
3  
4  
5 if __name__ == "__main__":  
6     print(hello_world())  
7
```

The Run and Debug icon (a play button with a bug) in the top right corner of the editor is highlighted with a red box. Below the editor, the TERMINAL panel is active, showing the command to run the script:

```
$ /bin/python /home/zachcran/workspaces/work/projects/2026_SIMCODES/repos/hello_world/hello.py  
Hello World!
```

The terminal output shows 'Hello World!' followed by a prompt. The status bar at the bottom indicates the file is at line 6, column 25, with 4 spaces, UTF-8 encoding, and LF line endings. The Python version is 3.14.5 64-bit.

Acknowledgements

- NSF for funding the REU.
- ISU and Ames National Lab



Installing Python (macOS)

- Open VSCode
 - At the top, select “Terminal”, “New Terminal”
 - Install UV with the following command:
`curl -LsSf https://astral.sh/uv/install.sh | sh`
 - Install Python:
`uv python install 3.14`
-
- Windows can also install UV at this point with:
`pip install --user uv`